

CMBSG Hardware Implementation and Development Process Guide

VOLUME 2

Mechanical Installation & Assembly Manual

REFERENCE DESIGN BASIS: 10,000 Nm³/HR CMBSG MODULE
DOCUMENT CLASS: ENGINEERING DELIVERABLE — CONSTRUCTION

Read Before Use

This volume assumes Volume 1 (Procurement, Vendor Specifications & BOM) equipment has been received, inspected against its Inspection & Test Plan, and released to site. All dimensions, torque values, and tolerances in this volume are for the corrected 10,000 Nm³/hr reference design basis established in Volume 1, Section 1. **Do not begin Stage 3 mesh tower assembly or Stage 4 hydraulic press assembly until the site-specific lifting plan and permit-to-work have been approved** — these two activities carry the highest mechanical risk in the entire installation (heavy lift + fragile titanium mesh; stored hydraulic energy). Sequence-critical steps are marked **HOLD POINT** throughout.

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1 How to Use This Volume

1.1 Scope

This volume covers mechanical installation and assembly only: rigging, setting, aligning, bolting, welding, and pressure-testing the equipment specified in Volume 1. It does **not** cover electrical/instrumentation termination (Volume 3), commissioning/startup (Volume 4), or process safety interlocking (Volume 5) — those follow after mechanical completion is certified in this volume's Section 11.

1.2 Reading Order

Follow Sections 3–6 in stage order (1→2→3→4) even though real-world construction often runs stages in parallel with multiple crews. Where a step in one stage has a hard dependency on another stage's completion (for example, Stage 4's curing tunnel foundation must cure before the press is set on it), this is called out explicitly with a **DEPENDENCY** note.

1.3 Hold Point Convention

Hold Point Convention

A **HOLD POINT** is a step that must not proceed without a specific inspection, approval, or permit being closed out first. Hold points in this volume are: (1) lifting plan approval before any lift >2 tonnes or >3m height, (2) pressure vessel hydrotest witness before insulation/lagging, (3) titanium mesh installation QA sign-off before tower closure, (4) hydraulic press factory-test certificate review before on-site pressurization. These mirror the ITP hold points defined in Volume 1, Appendix B.

2 Pre-Installation Requirements

2.1 Site Readiness Checklist

Table 1: Site readiness checklist — verify before equipment arrives.

Item	Acceptance Criteria
Foundations poured and cured	Minimum 28-day concrete cure (or per mix design early-strength certificate) before accepting static equipment loads
Foundation bolt templates verified	Survey check against vendor general arrangement (GA) drawings, tolerance $\pm 3\text{mm}$
Access roads / crane pad	Rated for heaviest single lift (see Section 2.2, rigging plan) plus crane outrigger bearing pressure
Utilities stubbed to battery limits	Cooling water, compressed air (instrument air, minimum Class 3 per ISO 8573-1), power, at each stage boundary
Laydown area	Minimum 200 m ² covered, humidity-controlled area for titanium mesh and enzyme cold-chain storage
Fire protection water available	Per local code, especially near HF/chemical storage (Volume 5)

2.2 Rigging and Lifting Plan Overview

Table 2: Major lift summary — confirm crane capacity before mobilization.

Item	Approx. Weight	Lift Height	Special Requirement
R-201 reactor (empty)	4.5–5.5 t	3–4 m set height	Glass lining — no chain contact with shell, use fabric slings only, vendor lifting lugs mandatory
T-101 sump tank	2.0–2.5 t	2 m	Standard 4-point lift
C-301 mesh tower sections	1.5–3 t per section	up to 12 m stacked	Tandem lift with tag lines ; assemble in 3–4 sections, mesh installed after shell erection (Section 5)
HP-401 hydraulic press	12–18 t	1–2 m onto foundation	Certified spreader bar, vendor rigging drawing mandatory, HOLD POINT for lift plan sign-off
PM-401 pug-mill	3–4 t	2 m	Standard 4-point lift

HOLD POINT — Lifting Plan

No lift exceeding 2 tonnes or 3 m height may proceed without a written, engineer-approved lifting plan identifying: crane make/model and load chart, rigging configuration, ground bearing pressure check, exclusion zone, and a named lift supervisor. This applies in full to the R-201 reactor (glass-lined, cannot be repaired if chipped), the C-301 tower sections (titanium mesh integrity), and the HP-401 press (18-tonne class lift).

2.3 Material Handling — Titanium Mesh and Enzyme Cold Chain

Special Handling Requirements

- **Titanium mesh screens (20/80/150 mesh, Ti Grade 2):** ship and store flat or on purpose-built spools, never folded. Handle with cotton gloves (bare-hand oils can affect the enzyme-immobilization surface prep done at the vendor). Store in the laydown area at <60% RH to prevent oxide layer growth beyond the vendor's controlled 5nm TiO₂ spec (see Volume 1, Section 4).
- **CA-KR1 enzyme:** arrives lyophilized at –20 °C. **Do not install into the dosing skid (DS-301) until Stage 3 electrical/instrumentation is complete and the cold-storage unit is commissioned and verified stable** (Volume 4). Maintain unbroken cold chain from delivery truck to storage freezer — log temperature excursions, anything >4 hours above –15 °C requires supplier consultation before use.

3 Stage 1 Mechanical Assembly — Gas Conditioning & Venturi Quench

3.1 Assembly Sequence Overview

1. Set and grout T-101 sump tank on foundation (Section 3.2)
2. Erect inlet ductwork D-101 from battery limit to venturi inlet flange (Section 3.3)
3. Set V-101 venturi assembly, mate to sump tank top flange (Section 3.4)
4. Install F-101 ID fan downstream of Stage 3 tie-in (fan is physically located after Stage 3 per original P&ID — confirm against site-specific P&ID before final placement)
5. Install P-101 sump recirculation pump and piping
6. Hydrostatic test T-101 and V-101 pressure boundary (Section 9)
7. Install ceramic throat liner (Section 3.4.2) — **after** hydrotest, before insulation

3.2 Sump Tank (T-101) Setting Procedure

Step-by-Step: T-101 Setting

1. Verify foundation bolt pattern against vendor GA drawing, $\pm 3\text{mm}$ tolerance on bolt circle.
2. Install leveling shims (stainless steel, matched to Super Duplex base ring to avoid galvanic coupling with carbon steel shims).
3. Lift per approved rigging plan; lower onto foundation bolts at $<0.1\text{ m/s}$ descent rate for the final 300mm.
4. Level tank to $\leq 2\text{ mm/m}$ across the base ring using precision level; check in two perpendicular axes.
5. Torque foundation bolts in a star pattern to the vendor's specified torque (typically M24 bolts at 450–500 N·m for this vessel class — confirm against vendor's certified drawing, do not use a generic torque chart for Super Duplex fasteners).
6. Verify nozzle orientations against P&ID before final bolt torque (nozzles cannot be easily re-oriented once grouted).
7. Grout base per vendor procedure; cure minimum 72 hours before applying any pipe loads to nozzles.

3.3 Inlet Ductwork (D-101) Installation

Table 3: D-101 ductwork installation tolerances.

Parameter	Requirement
Alignment	$\leq 3\text{mm/m}$ deviation from design centerline
Flange bolt-up	Cross-pattern (star) tightening sequence, minimum 3 passes to final torque
Gasket	Per vendor spec for saturated acidic gas service (PTFE-faced or equivalent per Volume 1 corrosion allowance basis)
Support spacing	Per vendor drawing; verify no support falls within 300mm of a weld seam
Expansion provision	Confirm expansion joint (if specified) is installed with correct pre-set for the $25 \rightarrow 180^\circ\text{C}$ thermal cycle

3.4 Venturi Assembly (V-101) Installation

3.4.1 Mechanical Setting

Step-by-Step: V-101 Setting

1. Confirm the throat section (280mm Ø) orientation matches the P&ID flow direction before lifting — venturi bodies are frequently asymmetric and can be installed backward if not checked.
2. Mate the venturi outlet flange to the T-101 top nozzle. Use new gaskets rated for the acidic condensate service (do not reuse test gaskets from any prior hydrotest).
3. Install the cooling-water quench nozzle per P&ID, verify spray pattern orientation (should atomize into the throat, not impinge directly on the ceramic liner in a way that could cause localized thermal shock).
4. Do not install the ceramic throat liner yet — see Section 3.4.2, liner goes in after hydrotest.

3.4.2 Ceramic Throat Liner Installation (Post-Hydrotest)

Note on Design Cut-Point

Volume 1 flagged that the original reference's claimed 32 μm aerodynamic cut-diameter is inconsistent with its own Calvert-model inputs (recomputed value $\approx 0.15 \mu\text{m}$). This does not change how the liner is installed, but it does change what you should expect when the vendor's own performance curve is checked during commissioning (Volume 4) — do not be alarmed if the commissioning team requests a finer-than-expected particle-size cut verification; this is expected given the correction, not a sign the liner was installed wrong.

Step-by-Step: Liner Installation

1. Confirm hydrotest (Section 9) is complete and the pressure boundary is signed off — **HOLD POINT**.
2. Clean throat bore of any weld spatter or hydrotest residue.
3. Install 95% Al_2O_3 ceramic liner segments per vendor's bonding procedure (typically a refractory castable or mechanically-retained tile system — follow vendor instructions exactly, thermal-shock cracking is the dominant liner failure mode and is almost always caused by incorrect cure/dry-out schedule).
4. Cure per vendor's specified heat-up schedule before any hot gas is introduced (typically a staged dry-out: ambient $\rightarrow 110^\circ\text{C} \rightarrow 300^\circ\text{C} \rightarrow$ operating temperature, each stage held for the vendor-specified soak time).
5. Visually inspect for hairline cracks after cure; tap-test (coin-tap method) for disbonded areas before accepting.

4 Stage 2 Mechanical Assembly — Matrix Blending Loop

4.1 Assembly Sequence Overview

1. Set R-201 leaching reactor (glass-lined — special handling, Section 4.2)
2. Set T-202 dissolution tank
3. Install MX-203 static mixer and interconnecting piping
4. Install P-202 recirculation pump (Hastelloy C-276)
5. Install slag hopper feeder and chitosan dissolution equipment
6. **NEW per Volume 1 correction:** install the SI (saturation index) trim dosing sub-loop on T-202 — see Section 4.4
7. Hydrotest R-201 and T-202 pressure boundaries

4.2 Glass-Lined Reactor (R-201) — Special Handling Procedure

Glass-Lined Equipment — Irreversible Damage Risk

Glass linings cannot be field-repaired to original integrity. A single chain-sling contact, dropped tool, or over-torqued nozzle bolt can chip the lining and create a corrosion initiation site that will fail in service. Follow the vendor's handling procedure exactly; where this section conflicts with a generic rigging practice, the vendor procedure governs.

Step-by-Step: R-201 Setting

1. Inspect glass lining visually and by holiday (spark) test per vendor procedure **before** lifting off the delivery truck — establish a damage baseline before your crew touches it.
2. Lift using only the vendor-supplied lifting lugs and fabric (never chain) slings; no sling shall contact the vessel shell directly.
3. Lower onto foundation at <0.05 m/s for the final 150mm — slower than the T-101 sump tank setting, because impact tolerance is much lower.
4. Do not weld anything to the reactor shell or supports after the glass-lining process — all nozzle connections must be bolted, per vendor design.
5. Re-run the holiday test after setting and before any piping is connected, to confirm no transit or lifting damage occurred.
6. Torque nozzle flange bolts to vendor spec using a calibrated torque wrench in a star pattern, in a minimum of 3 progressively-increasing passes — over-torquing a glass-lined nozzle flange is a common cause of lining cracks at the nozzle root.

4.3 Dissolution Tank (T-202) and Piping

Table 4: T-202 and Stage 2 piping installation notes.

Item	Requirement
T-202 setting	Standard 316L SS vessel setting per Section 3.2 procedure (sump tank), adjusted for T-202's baffled internals — verify baffle orientation against vendor drawing before final bolt-up (baffles are sometimes shipped detached for transport)
MX-203 static mixer	Install with the correct flow-direction arrow visible; verify no reducers/expanders were substituted for the specified DN80 CPVC line size, since the mixer's helical elements are sized for a specific velocity range
P-202 pump	Baseplate grouted and leveled to ≤ 0.05 mm/m before coupling alignment (see Section 8, alignment tolerances); Hastelloy C-276 wetted parts — verify material certs match Volume 1 spec at receipt, before installation

4.4 SI Trim Sub-Loop Installation (Volume 1 Correction)

Why This Sub-Loop Exists

Volume 1, Section 4.2 found that the original reference's saturation index calculation (claimed -0.58) recomputes to $+0.42$ from its own stated inputs — meaning the nominal chemistry runs closer to supersaturation (premature CaCO_3 scaling risk in T-202) than the original design assumed. This sub-loop is the mechanical answer to that finding: an independent, fast-acting acid trim capable of pulling the SI back down without waiting on the main leaching reactor's slower response.

Step-by-Step: SI Trim Sub-Loop Installation

1. Install a dedicated metering pump (0–5 L/hr, 316L SS wetted parts) on a new nozzle on T-202 — confirm with the vessel vendor whether this requires a re-rated nozzle schedule or can use a spare nozzle from the original design (many vendors include 1–2 spares by default; verify before drilling a new opening in a pressure vessel, which would void the original certification).
2. Route dilute acetic acid supply from the existing Stage 2 acid dosing header (shared with R-201, per original P&ID) via a dedicated branch with its own isolation valve.
3. Locate the injection point downstream of the main agitator discharge for rapid mixing, not near the tank wall.
4. Tag this loop distinctly (e.g., “AIC-202-TRIM”) in the P&ID handed to Volume 3 (E&I) so the control narrative correctly treats it as a fast trim loop cascaded from the pH/conductivity soft-sensor, not the main pH control loop.

5 Stage 3 Mechanical Assembly — Kinetic Mesh Absorber Column

5.1 Assembly Sequence Overview — Highest-Risk Stage

Why Stage 3 Gets the Most Detail

This stage combines the tallest structure (12m), the single most expensive and fragile component (titanium mesh, 16–20 week lead time per Volume 1), and the open engineering question flagged in Volume 1 (Hatta number / enhancement factor). A mesh-installation error here is expensive to fix (requires re-opening a closed tower) and a design change from the pending pilot verification could require additional tower height. Sequence this stage carefully.

1. Erect C-301 tower shell in sections (typically 3–4 field welds/bolted flanges for a 12m tower, per vendor transport sectioning)

2. Install internal support rings for the three mesh tiers **before** final section stack-up
3. **HOLD POINT:** verify tower plumbness (vertical alignment) before mesh installation
4. Install Mesh Layer 1 (top, 20 mesh, coarse)
5. Install Mesh Layer 2 (middle, 80 mesh)
6. Install Mesh Layer 3 (bottom, 150 mesh, fine)
7. Install ultrasonic transducers US-301/2/3 at each tier
8. Install matrix distribution header (trickle-bed feed) above Layer 1
9. Close tower, install access ports/manways
10. Hydrotest / pneumatic leak test per Section 9
11. Install DS-301 enzyme dosing skid (mechanical only — do not charge with enzyme yet, see Section 2.3)

5.2 Tower Shell Erection and Plumbness Verification

Step-by-Step: Tower Erection

1. Set base section on foundation, verify level to $\leq 1\text{mm/m}$ (tighter than the sump tank tolerance, because a 12m tall structure amplifies base tilt into significant top-of-tower deflection).
2. Stack subsequent sections per vendor match-marking; verify flange orientation against the P&ID at each joint (a 90° rotation error at any joint will misalign the internal mesh support rings).
3. After full stack-up, verify plumbness using a theodolite or laser plumb from base to top: total deviation must not exceed 1:500 of tower height (24mm over 12m) per standard tall-vessel erection practice.
4. **HOLD POINT:** plumbness verification must be signed off by the erection engineer before mesh installation begins — installing delicate titanium mesh into an out-of-plumb tower risks uneven matrix distribution and premature mesh fatigue at points of uneven mechanical loading.

5.3 Titanium Mesh Installation Procedure

Open Engineering Item — Read Before Proceeding

Volume 1 flagged that the Hatta number computed from the original reference's own stated inputs is 0.088, not the claimed 0.877 (a $10\times$ decimal error), and that the resulting mass-transfer enhancement factor is closer to 1.0 than the claimed 1.25. **This means the tower height (12m nominal, Section 5.2 of Volume 1) may need to increase if the pilot verification (Volume 4) shows lower capture efficiency than the original 95% target at the specified enzyme loading and residence time. Recommendation:** install the mesh support rings and tower internals with provision for a bolt-on extension section (an additional 2–4m of shell with matching internal support rings, procured but not necessarily installed) so that if the pilot result requires more residence time, the tower can be extended without full disassembly. Discuss this contingency with the vendor before finalizing the tower shell purchase order.

Step-by-Step: Mesh Installation — Each Tier

1. Unroll mesh from its shipping spool onto a clean, padded work surface — never drag across a rough or metallic surface, which can create burrs that snag on wire during tensioning.
2. Inspect for shipping damage: creases, torn strands, discoloration (which may indicate an oxide layer thicker than the vendor's controlled 5nm spec, potentially affecting the CA-KR1 enzyme immobilization performed at the vendor).
3. Cut to the internal support ring dimension with a 10–15mm overlap margin for clamping, using vendor-approved cutting method (typically fine abrasive wheel; avoid heat-generating methods that could anneal the wire and change its fatigue properties).
4. Position on the support ring and tension evenly using a star pattern (similar principle to bolt torque sequencing) to avoid a wrinkle or uneven sag that would create preferential flow channeling.
5. Clamp per vendor-specified clamp ring torque; re-check tension after 24 hours (titanium mesh can relax slightly after initial tensioning).
6. Repeat for all three tiers, working top-down (Layer 1 first, since it's installed highest in the tower and subsequent work platforms are removed as you descend).
7. **QA sign-off required for each tier before proceeding to the next** — document mesh lot number, installer, tension reading, and visual inspection result per tier.

5.4 Ultrasonic Transducer Mounting

Table 5: US-301/2/3 mounting requirements (corrected amplitude basis per Volume 1).

Parameter	Requirement
Mounting location	One transducer set per mesh tier, positioned per vendor's coupling-efficiency drawing (typically edge-mounted on the support ring, not directly on the mesh itself)
Acoustic coupling	Follow vendor's coupling compound or direct-bond procedure; air gaps at the coupling interface will drastically reduce delivered amplitude
Corrected amplitude target	0.32 μm minimum required (corrected from originally-stated 0.7 μm , see Volume 1 Section 5.3); specified drive amplitude 3.5 μm gives $\sim 10.9\times$ margin — flag to vendor that this may be over-specified; request vendor confirm no premature mesh fatigue risk from excess amplitude before final commissioning
Cabling	Route transducer cabling away from any HF or NaBH_3CN handling area (Volume 5) to avoid corrosive atmosphere exposure to cable jacketing

6 Stage 4 Mechanical Assembly — Consolidation and Press Foundry

6.1 Assembly Sequence Overview

1. Confirm curing tunnel (CT-401) foundation has cured minimum 28 days (**DEPENDENCY: do not set HP-401 press until this foundation is verified, since press vibration during test-run could affect adjacent green concrete**)
2. Set PM-401 pug-mill
3. Set HC-401 hydrocyclone
4. Set HP-401 hydraulic press — **HOLD POINT**, see Section 6.3
5. Erect CT-401 curing tunnel structure
6. Install interconnecting piping/conveyance between PM-401 $\rightarrow$ HC-401 $\rightarrow$ HP-401 $\rightarrow$ CT-401
7. Hydrotest/pressure-test HP-401 hydraulic circuit per vendor factory test certificate cross-check

6.2 Pug-Mill and Hydrocyclone Setting

Table 6: PM-401 / HC-401 setting notes.

Item	Requirement
PM-401 foundation	Verify vibration isolation pads (if specified by vendor) are installed before setting — twin-shaft pug-mills generate significant dynamic loading
PM-401 shaft alignment	Confirm both shafts rotate freely by hand before motor coupling, verify no binding from Hardox blade clearance to housing (typically 3–5mm per vendor spec)
HC-401 orientation	Apex angle and orientation per vendor drawing — incorrect orientation significantly degrades underflow solids concentration

6.3 Hydraulic Press (HP-401) Setting — Hold Point Procedure

HOLD POINT — Hydraulic Press Installation

The HP-401 press stores significant hydraulic energy at its 200 bar operating pressure with a 175 bar nitrogen pre-charge accumulator. **Do not connect site hydraulic power or attempt any pressurization until:** (1) the vendor's Factory Acceptance Test (FAT) certificate has been reviewed and matches the as-shipped unit's serial number, (2) all guarding (light curtains, Category 3 PLd per ISO 13849) specified in Volume 1 is installed and verified, (3) the relief valve (set at 220 bar per Volume 1) has its own calibration certificate on file. This is detailed further in Volume 5 (Safety Systems).

Step-by-Step: HP-401 Setting

1.

Confirm foundation reinforcement matches the vendor’s dynamic loading specification (250t press class, verify against structural drawing — this is not a standard equipment pad, it requires impact-factor design per Volume 1, Section 10).

2.

Rig per the approved lifting plan (Section 2.2, Table 2) using the vendor’s certified spreader bar — do not improvise rigging on an 18-tonne press.

3.

Lower onto foundation bolts, verify level to $\leq 0.5\text{mm/m}$ across the press bed (tighter tolerance than static vessels, since die-cavity alignment across all 4 cavities depends on a level, rigid base).

4.

Torque foundation bolts per vendor spec; grout per vendor procedure.

5.

Connect hydraulic power unit (HPU) per vendor’s piping and instrumentation drawing — flush all hydraulic lines per vendor’s cleanliness spec (typically ISO 4406 code specified by the vendor) before first pressurization, to avoid introducing contamination that could damage precision seals.

6.

Do **not** energize until Volume 5 safety system verification (light curtains, relief valve calibration, E-stop function) is complete and documented.

6.4 Curing Tunnel (CT-401) Erection

Table 7: CT-401 erection notes.

Item	Requirement
Structure	Insulated steel shell sections, bolted per vendor drawing; verify liner (PVC or FRP) seams are sealed per vendor procedure — gaps allow humidity loss, compromising the 90% RH cure target
Conveyor/tray system	Align per vendor drawing; verify no binding across the full 18m length before first production trial
Cure-time note (Volume 1 correction)	The original reference’s Arrhenius rate constant for pozzolanic curing contained an arithmetic error (stated 0.013 hr^{-1} vs. correct 0.082 hr^{-1}). This doesn’t change tunnel installation, but flag to the commissioning team (Volume 4) that a cure-time sensitivity trial (8h/10h/12h) is recommended rather than assuming 12 hours is the minimum required residence time.

7 General Piping Installation Requirements

Table 8: General piping installation requirements, all stages.

Requirement	Detail
Flange bolt-up sequence	Star pattern, minimum 3 passes, final pass to full torque; document torque values on a bolt-up record sheet per joint
Gasket selection	Match to Volume 1 corrosion allowance and service pH/temperature; never substitute without engineering approval
Pipe supports	Per vendor / design drawing spacing; verify no support welded directly to a lined or coated pipe without vendor approval
Cleanliness	Blow down / flush all lines before final tie-in, particularly CPVC lines (Stage 2) which are susceptible to internal debris causing static mixer fouling
Dissimilar metal joints	Use appropriate isolation (dielectric unions or equivalent) between Super Duplex, 316L, Hastelloy, and carbon steel sections to avoid galvanic corrosion

8 Welding Procedures and NDT Requirements

Table 9: Welding and NDT requirements by joint category.

Joint Category	Welding Process / Procedure	NDT Requirement
Pressure vessel shell (T-101, R-201*)	Per vendor WPS (Welding Procedure Specification), qualified to ASME IX or equivalent	100% RT (radiographic) on longitudinal seams, per Volume 1 Appendix B ITP
Ductwork (D-101)	GMAW or per fabricator standard	Visual + spot UT per fabricator QC plan
Structural steel (tower supports, press foundation embeds)	Per AWS D1.1 or local structural code	Visual, MT (magnetic particle) on critical connections
Field piping welds	Per ASME B31.3	RT or UT per line-class requirement; 100% for Category M (hazardous) service lines

**R-201 is glass-lined and shipped fully welded from the vendor — no field welding is performed on this vessel; this row is included for completeness of the vendor's own QC record review.*

9 Pressure Testing Procedures

Standard Hydrostatic Test Procedure

1. Isolate the system under test; install test blinds at all open connections per the test package boundary drawing.
2. Fill with water (or vendor-approved test fluid — note some vessels, e.g. glass-lined R-201, may require a specific test fluid per vendor to avoid lining damage from test water chemistry).
3. Vent all high points to ensure no air pocket remains (air compressibility invalidates the test and is a safety hazard).
4. Pressurize slowly to $1.5 \times$ design pressure (standard ASME VIII Div. 1 hydrotest factor) at a controlled rate per code, typically no faster than 10% of test pressure per minute.
5. Hold for minimum 30 minutes (or per governing code / vendor spec) — inspect all welds, flanges, and nozzles for leaks.
6. Record test pressure, hold time, temperature, and inspector sign-off on the test certificate.
7. Depressurize slowly; drain and dry per vendor procedure (critical for glass-lined and Ti-mesh-containing equipment to avoid residual moisture issues).

Special Case — C-301 Mesh Tower

Because titanium mesh is installed **before** the tower is closed (Section 5.3–5.4), the pressure/leak test for C-301 should be performed as a pneumatic leak test (soap-solution or equivalent) at a much lower test pressure (typically $1.1 \times$ operating pressure, not $1.5 \times$ design, and only after consulting the mesh vendor on whether pneumatic pressure cycling could affect mesh tensioning) rather than a full hydrostatic test that would submerge the enzyme-coated mesh in test water unnecessarily.

10 Alignment and Leveling Tolerances

Table 10: Alignment and leveling tolerances by equipment class.

Equipment	Tolerance	Method
Static vessels (T-101, R-201, T-202)	$\leq 2\text{mm/m}$	Precision level, two perpendicular axes
C-301 tower plumbness	1:500 of height (24mm over 12m)	Theodolite / laser plumb
HP-401 press bed	$\leq 0.5\text{mm/m}$	Precision level across bed
Pump/motor coupling alignment	$\leq 0.05\text{mm}$ parallel, $\leq 0.05\text{mm}/100\text{mm}$ angular	Dial indicator or laser alignment tool

11 Mechanical Completion Checklist

Table 11: Mechanical completion checklist — sign-off required before handover to Volume 3 (E&I).

Item	Verification
All equipment set, leveled/aligned per Section 8	Survey/alignment records on file
All pressure boundaries hydrotested per Section 9	Test certificates on file, signed by inspector
All welds NDT'd per Section 7	NDT reports on file, zero open defects
Titanium mesh installed, tensioned, QA-signed per tier	Mesh installation log complete (Section 5.3)
Ceramic throat liner cured and inspected	Liner inspection report on file (Section 3.4.2)
Glass lining holiday-tested post-installation	Holiday test report on file (Section 4.2)
HP-401 FAT certificate reviewed and matched to serial number	FAT certificate on file (Section 6.3)
All bolted joints torqued and recorded	Bolt-up records on file (Section 7)
Punch list items closed or formally deferred with engineering sign-off	Punch list document on file

Handover to Volume 3

Mechanical completion as defined by this checklist is the gate for beginning electrical and instrumentation work (Volume 3). Partial mechanical completion (e.g., Stage 1 and 2 complete, Stage 3 pending mesh delivery) is acceptable for a phased handover provided each stage's own checklist is fully closed before that stage's E&I work begins.

A Appendix A: Generic Torque Chart Reference

Use Vendor Torque Values When Available

The values below are generic reference torques for standard fastener grades and are provided only for items where Volume 1 vendor datasheets do not specify a torque value. **Always use the vendor's certified torque specification when one exists** — this is especially critical for glass-lined equipment (R-201) and the hydraulic press (HP-401), where generic values are not appropriate.

Table 12: Generic bolt torque reference (Grade 8.8 carbon steel, lubricated, for non-critical structural connections only).

Bolt Size	Torque (dry)	Torque (lubricated)	Application Note
M12	86 N·m	65 N·m	Instrument brackets, light structural
M16	210 N·m	155 N·m	General piping flanges
M20	425 N·m	315 N·m	Medium structural connections
M24	730 N·m	545 N·m	Heavy structural, vessel foundation (confirm against vendor spec)

B Appendix B: Bolt-Up Record Sheet Template

Joint ID	Bolt Size	Target Torque	Achieved Torque	Technician / Date

Table 13: Blank bolt-up record template — one row per flange/joint, retain as part of mechanical completion dossier.

C Appendix C: Lifting Plan Template

Field	Entry
Item to be lifted	
Weight (confirmed, not estimated)	
Crane make/model & load chart reference	
Rigging configuration	
Ground bearing pressure check	
Exclusion zone radius	
Named lift supervisor	
Weather limits (wind speed, etc.)	
Approval signature & date	

Table 14: Blank lifting plan template for lifts exceeding 2 tonnes or 3m height.

D Appendix D: Cross-Reference to Volume 1 Corrections

Volume 1 Finding	Mechanical Impact (This Volume)	Section
Calvert d_{50} inconsistent	None to installation procedure; expect different commissioning result	Section 3.4.2
Saturation index sign error	New SI trim sub-loop installed on T-202	Section 4.4
Hatta number / enhancement factor unsupported	Tower built with bolt-on extension provision in case pilot requires more residence time	Section 5.3
Titanium property inconsistency	Corrected amplitude spec ($0.32\ \mu\text{m}$) used for transducer mounting	Section 5.4
Arrhenius rate constant error	Cure-time sensitivity trial recommended, no installation change	Section 6.4
Stage 1 ΔP off by $10\times$	No mechanical impact; SIS setpoint unaffected	N/A (Volume 5)